

Physically-Motivated Compound-Degradation Image Restoration

KLA AI Hackathon, IITH'26

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Problem Statement & Dataset Overview

- **Objective:** Develop a robust model to reconstruct high-quality images from degraded, noisy, low-resolution (`.npy`) inputs.
- **Challenge:** Real-world physical degradation is complex, requiring models to handle **compound corruptions** simultaneously (Speckle + AWGN + Downsampling).
- **Dataset:** Supervised pairs of degraded $\leftrightarrow$ Ground Truth (256×256). Test set includes out-of-distribution samples to evaluate generalization.
- **Evaluation:** Joint optimization of MSE, PSNR, SSIM, LPIPS + Inference speed.

Noisy Image



Ground Truth Image



Motivation behind EDA

- **Objective:** Statistically predict the image degradation process to understand its effect on the possible manifold space, serving as an additional supervision prior for the restoration task.
- **Context:** We do not know the exact noise distribution of the images used during the NAFNet pretraining phase.
- **Hypothesis:** By embedding the physical behavior of our specific case (Down-sampling + Speckle + AWGN) into the finetuning process, the model's performance can be significantly improved.
- **Role of EDA:** Rather than perfectly estimating deterministic parameters, EDA helps us identify the most likely random perturbations from possible options and estimate their underlying parameter ranges.

Spatial Match & Proxy Downsampling

- **Scale Factor:** The noisy images are downsampled by a factor of 2×2 compared to the ground truth.
- **Proxy Downsampling:** Evaluated Nearest Neighbor vs. Block Mean.
- **Selection:** The `block_mean` approach is the optimal proxy due to a significantly lower residual MSE (0.0081 vs 0.0104).

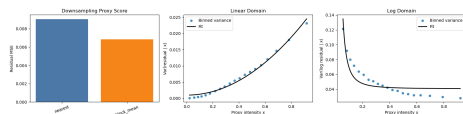


Figure: Downsampling Proxy Comparison

Noise Modeling: Speckle & Additive

- **Linear Model:** $\text{Var}(r|x) \approx a \cdot x^2 + b$
 - Speckle noise ($\sigma \approx 0.167$)
 - Additive noise ($\sigma \approx 0.031$)
- **Log-Domain & PCA:** Log residual analysis reveals a significant $1/x^2$ term. PCA on linear residuals cross-checks a baseline variance ($\sigma \approx 0.021$).
- **Implication:** The noise is a complex mix of multiplicative and additive components.

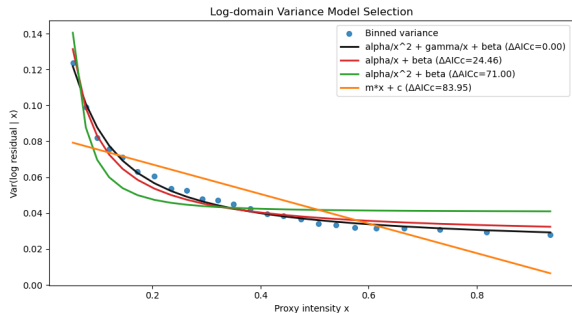


Figure: Noise Variance Model Fit

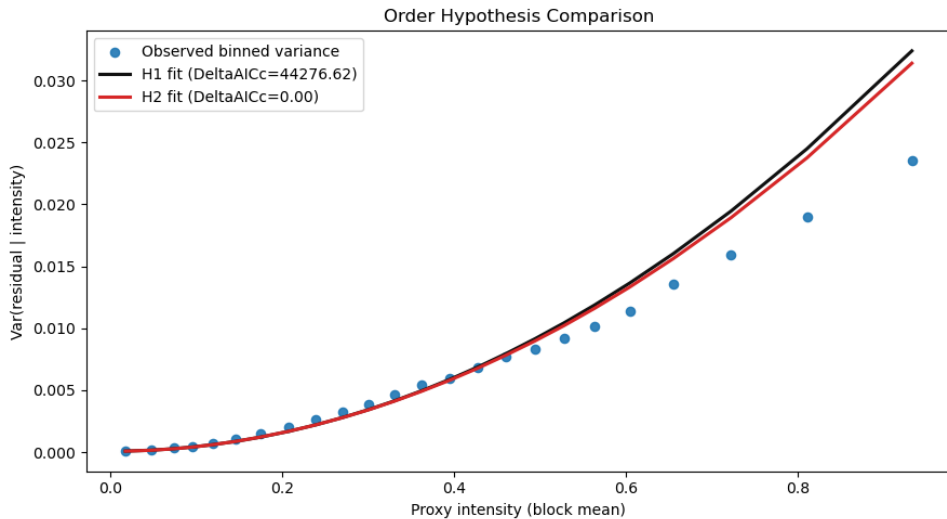
Degradation Pipeline Hypothesis

- **Goal:** Determine the chronological sequence of degradation.
- **Hypotheses Tested:**
 - **H1:** Downsample $\rightarrow$ Speckle $\rightarrow$ Additive
 - **H2:** Speckle $\rightarrow$ Downsample $\rightarrow$ Additive
- **Results:** AICc ranking strongly favors **H2** ($\Delta AICc = 0.00$). H1 incurs a heavy statistical penalty.

Final Inferred Pipeline:

Ground Truth $\rightarrow$ Speckle Noise $\rightarrow$ Block Mean Downsample $\rightarrow$ AWGN

Hypothesis Comparison



NAFNet: Nonlinear Activation Free Network

- **Architecture:** Single-stage U-Net with dense skip connections (low inter-block complexity).
- **Critical Innovation:** Removes standard nonlinear activations (ReLU, GELU, Sigmoid) entirely.

- **SimpleGate:** Replaces GELU with element-wise multiplication of channel-split features:

$$Y = X_1 \odot X_2$$

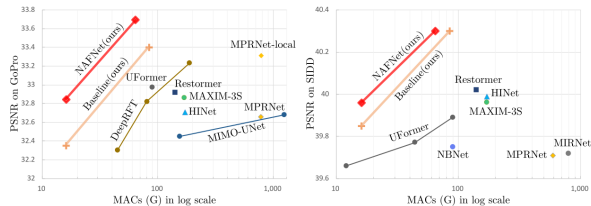
Nonlinearity is introduced purely via gating, mimicking GLU without extra activations.

- **Simplified Channel Attention (SCA):** Removes ReLU/Sigmoid from CA. Uses global average pooling and a linear projection to reweight channels efficiently.
- **Result:** A “nonlinear activation-free” network that simplifies the computational graph, stabilizes training, and drastically reduces MACs.

Why NAFNet Revolutionized Efficient Image Restoration

- **State-of-the-Art Performance:**
Achieves ~ 40.30 dB PSNR on SIDD and ~ 33.69 dB on GoPro benchmarks.
- **Computational Efficiency:**
 - $< 50\%$ MACs of Restormer (Denoising).
 - Only 8.4% MACs of MPRNet (Deblurring).
- **Why it works:** Replaces expensive self-attention maps and multi-stage pipelines with simple convolutions + element-wise gates.
- **Hackathon Fit:** High efficiency enables rapid inference optimization and fits within strict compute constraints while maximizing SSIM/LPIPS.

PSNR vs. MACs (SIDD Benchmark)



Our Implementation: NAFNetH2SR for Joint SR & Denoising

Adapting NAFNet to our Compound Degradation Challenge:

- **Task:** Joint Speckle/AWGN Denoising + Deblurring + $2\times$ Super-Resolution.
- **Grayscale Handling Pipeline:**
 - ① Upsample LR input ($1 \times H \times W$) via bicubic interpolation.
 - ② **Replicate to 3 channels** for compatibility with official RGB NAFNet backbone.
 - ③ Restore via NAFNet $\rightarrow$ **Collapse to grayscale** via channel-wise mean: $\hat{y} = \frac{1}{3} \sum_{c=1}^3 \hat{x}_c$.
- **Advantages for Hackathon:**
 - **Robustness:** U-Net structure captures fine-grained noise removal while preserving structural coherence (SSIM/LPIPS friendly).
 - **Inference Speed:** Lightweight blocks enable fast processing of large test batches.
 - **Output:** Direct 256×256 float32 grayscale predictions ready for `.npy` submission.

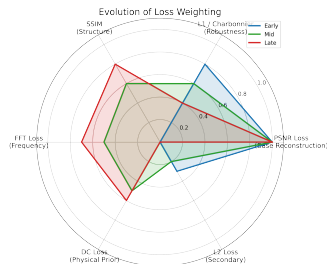
Objective Formulation: Multi-Faceted Loss

To prevent over-smoothing and enforce structural fidelity, we employ a dynamically weighted loss landscape:

- **Reconstruction (Pixel):** Official NAFNet PSNR Loss (log-scaled MSE) + Charbonnier Loss ($\sqrt{(x - \hat{x})^2 + \epsilon^2}$) for outlier robustness.
- **Perceptual & Structural:**
 - **SSIM Loss:** Enforces local structural similarity.
 - **LPIPS:** Learned perceptual feature matching.
- **Heteroscedastic Data Consistency (DC):**

$$\mathcal{L}_{DC} = \frac{1}{N} \sum \log \left(1 + \frac{(y - \mu)^2}{\nu \cdot \nu} \right)$$

A heavy-tailed Student-t formulation bridging the LR input (y) and downsampled prediction (μ), isolated from variance collapse.



Weight of the different loss components

Training Dynamics: OOD Augmentation & Curriculum

Goal: Force the network to learn robust feature extraction rather than memorizing the exact noise distribution.

- **Synthetic OOD Degradations:**

- *Photometric:* Affine shifts, dynamic Gamma correction.
- *Noise:* Speckle, Poisson, Gaussian combinations.
- *Structural Defects:* 3x3 reflection blur, impulse noise, and spatial cutouts simulating sensor failure.

- **Curriculum Learning Schedule:**

- **Phase 1 (Robustness):** High probability ($p = 0.95$) and max transforms ($n = 5$). Early layers frozen to protect pre-trained features.
- **Phase 2 (Finetuning):** Transform probability and intensity linearly decay.
- **Phase 3 (Convergence):** Network fully un-frozen, evaluated primarily on pure dataset.

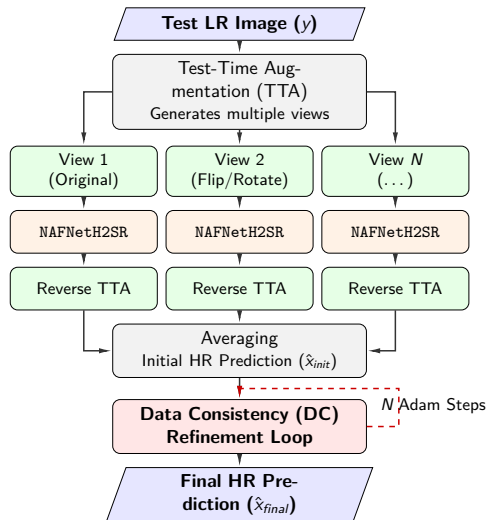
Inference Optimization: Maximizing the Test Score

• Test-Time Augmentation (TTA):

- *Balanced mode*: 4-way flips.
- *High mode*: 8-way (flips + rotations).
- Averages predictions across spatial transformations to reduce variance.

• Test-Time Refinement:

- Instantiates a local Adam optimizer at inference.
- Performs N gradient steps directly on the predicted HR output to minimize the Data Consistency (DC) loss against the specific test LR image.
- *Result*: Physically "locks in" the prediction to the exact noise profile of the test sample.



Quantitative Results & Leaderboard Performance

Local Validation (2% Held-Out):

- **PSNR:** 28.5 dB
- **SSIM:** 0.77

Kaggle Competition Results:

Phase	Score	Rank
Public LB	0.88614	12
Private LB	0.88303	10

OOD Generalization Hypothesis

Our model maintained high stability and actually **climbed two ranks** on the Private Leaderboard.

We hypothesize this resilience is directly attributable to our **heavy, curriculum-based OOD data augmentations**. By simulating extreme structural defects and photometric shifts during training, the model generalized perfectly to the unseen, severe degradations in the private test set.

Denoised Images (Randomly Sampled from Noisy Test Set)

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Sample 000000



Restored (256x256)



000079

Sample 000079

Noisy LR (128x128)



Restored (256x256)



000159

Sample 000159

Noisy LR (128x128)



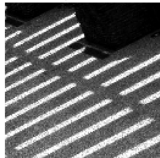
Restored (256x256)



000239

Sample 000239

Noisy LR (128x128)



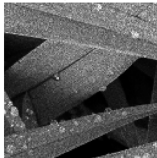
Restored (256x256)



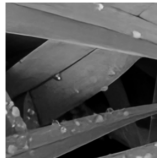
000319

Sample 000319

Noisy LR (128x128)



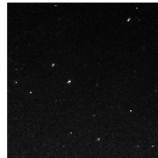
Restored (256x256)



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Sample 000399

Noisy LR (128x128)



Restored (256x256)

